Yongda Zhu, Ph.D.
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RESEARCH PROFILE

I study the **phase transition** of the Universe during cosmic **reionization** and how **galaxies** and **quasars** exchange mass, metals, and ionizing radiation with the **intergalactic medium**, which contains most of the Universe's baryonic matter. I combine quasar absorption spectroscopy, **JWST**, **ALMA**, **Keck**, **Roman**, and cosmological simulations to connect galactic-scale physics with the evolution of the **cosmic baryon cycle**.

My work has constrained the **timing and topology of late reionization**, established links between IGM transmission and galaxy environment, uncovered signatures of quasar feedback on intergalactic scales, revealed **metal-enriched baryon cycling** only 500 Myr after the Big Bang, and developed **AI-enabled imaging approaches** to studying the baryon cycle with Roman and JWST.

Selected record: Ph.D. in Physics (**2024**); **16** first-author papers, including publications in *ApJ*, *ApJL*, *Nature Astronomy*, and *MNRAS*; **3,500+** total citations (**430+** first-author citations); **3** student-led papers; PI/co-PI on **4** awarded Roman programs totaling **$1.2M** (**$0.6M** in direct PI support and **$0.6M** supporting collaborators through co-PI programs); PI of ALMA programs totaling **>40 hr** and more than **30 nights** on ground-based telescopes, including Keck and MMT; data or analysis leadership across approximately **2,000 hr** of competitively allocated JWST observations.

ACADEMIC APPOINTMENTS

- **JASPER Postdoctoral Scholar**, University of Arizona Aug. 2025 – Oct. 2026
  Mentors: Prof. Eiichi Egami & Prof. Xiaohui Fan
  Leading NIRCam/WFSS analysis for SAPPHIRES and studying gas–galaxy interactions across scales with JWST and machine learning.
- **Postdoctoral Research Associate**, NIRCam contract Feb. 2024 – Aug. 2025
  Mentors: Prof. Marcia Rieke & Prof. George Rieke
  Member of JWST NIRCam Science Team and MIRI U.S. Team.
  Led a systematic search for galaxies with extended line emission and candidate outflows using deep NIRCam medium-band imaging and machine learning.
  Leading the SMILES (JWST-GTO-1207) NIRSpec public data release and multi-band NIRCam photometric analysis of galaxies.
  Leading reduction and analysis of JWST/NIRSpec MSA spectroscopy of cosmic-noon galaxies and obscured AGN.
- **Graduate Student Researcher**, UC Riverside Jul. 2019 – Feb. 2024
  Mentor: Prof. George Becker
  Led Keck and ALMA observations of the IGM via high-z quasars; modeled the Lyα and Lyβ forest using cosmological simulations.

Provided observational evidence for a late and spatially inhomogeneous end to reionization.

PROFESSIONAL SERVICE

- **Grant Reviewer**: NSF Astronomy and Astrophysics Research Grants Review Panel 2026
- **Peer Reviewer**: *The Astrophysical Journal*, *ApJ Letters*, *JCAP*, and *Open Research Europe*; ALMA and Gemini observing proposals

EDUCATION

Ph.D. in Physics, University of California, Riverside Mar. 2024
Constraining the IGM during the Later Stages of Reionization using QSO Spectra; Advisor: Prof. George Becker

M.Sc. in Physics, University of California, Riverside Dec. 2019

B.Sc. in Astronomy, University of Science and Technology of China Jun. 2018
Testing Gravity Theories on Galactic Scales; Advisor: Prof. Xiao-Bo Dong

EXTERNAL FUNDING & PI PROGRAMS

Funding summary. Four awarded Roman Cycle 1 programs total \$1.20M: \$0.60M from two PI programs provides direct research support for Yongda Zhu, while \$0.60M from two co-PI programs supports collaborators. PI of two ALMA programs and three MMT/Binospec IFU programs; science and observing lead for repeated Keck campaigns and major contributor to JWST GO 4092.

Principal-Investigator Observing Programs

- **Roman 10995** — *Establishing the Post-Reionization Lyα Transmission–Environment Baseline*. **PI: Yongda Zhu**; \$150,000 awarded as direct PI research support.
- **Roman 19049** — *From Environment to Assembly: Roman Spatially Resolved Stellar Mass and SFH Maps*. **PI: Yongda Zhu**; \$450,000 awarded as direct PI research support.
- **Roman 19054** — *Do Filaments Influence Galaxies Beyond Local Environment? A Roman Census of Galaxy–Filament Connectivity*. **co-PI: Yongda Zhu**; \$450,000 awarded to support collaborators.
- **Roman 19029** — *Linking Galaxies to Their Neutral Gas Reservoir: A Roman–Rubin Synergy Survey of DLA Host Galaxies at Cosmic Noon*. **co-PI: Yongda Zhu**; \$150,000 awarded to support collaborators.
- **MMT/Binospec IFU 2026A** — *IFU Observations of a $z = 5.63$ Quasar–Companion System: Lyα Escape and UV Counterpart*. **PI: Yongda Zhu**.
- **MMT/Binospec IFU 2025B** — *Spatially Resolved Lyα Spectroscopy of a $z = 5.5$ Quasar with Strong Cold Outflows*. **PI: Yongda Zhu**.
- **ALMA 2024.1.00555.S (Cycle 11)** — *Galaxy Over/Underdensities Around IGM Transmission at $z = 5.7$: A Robust Constraint on Reionization*. **PI: Yongda Zhu**.
- **MMT/Binospec IFU 2024B** — *Ionization and Enrichment in the Reionization Epoch: A Pilot Study with Binospec IFU*. **PI: Yongda Zhu**.
- **ALMA 2022.1.00662.S (Cycle 9)** — *The Mean Free Path of Ionizing Photons at $z = 5.6$:*

A Robust Constraint on Reionization. **PI: Yongda Zhu**.

Observing Programs Led as a Ph.D. Student at UC Riverside

I led or published the science case and observing program for each allocation below. UC students could not serve as Keck PIs.

- JWST 4092 – *How Does Reionization End? A Search for [O III] Emitters in the Most Transparent Regions of the IGM Near Redshift Six*. Formal PI: George Becker.
  Zhu, et al. 2025, arXiv:2510.09568
- Keck/ESI [2021B_U036, 2022A_U035, 2022B_U042, 2023B_U049, 2024A_U281]: *The Mean Free Path at $z = 5.6$: Insights into Ultra-Late Reionization*.
- Keck/ESI [2019A_U014, 2020A_U121, 2021A_U039]: *Giant Lyα Troughs at $z < 6$: A Signature of Very Late Reionization?*.
  Zhu, et al. 2021, ApJ, 923, 223; **Zhu**, et al. 2022, ApJ, 932, 76
- Keck/LRIS [2019A_U147, 2019B_U147]: *The Mean Free Path at $z = 5$: A Key Constraint on Reionization Models*.
  Becker et al. 2021; **Zhu**, et al. 2023, ApJ, 955, 115
- Other allocations include: Keck/ESI [2021A, 2021B], Keck/DEIMOS [2020A], Keck/LRIS [2023B], Subaru/HSC [2020B, 2021A, 2021B, 2023B]

Selected Awards and Fellowships

- NSF/NRAO Student Observing Support Award ($40,000) 2023
- UCR Dissertation Year Fellowship ($7,200) 2022
- Benjamin C. Shen Memorial Award for Outstanding First-Year Graduate Researcher 2019
- Dean's Distinguished Fellowship, UC Riverside 2018
- Undergraduate research awards and scholarships, USTC 2015–2018
  CAS Undergraduate Research Award, NSFC Talented Students in Basic Sciences Program, CAS Innovation Training Program, and Outstanding Student Scholarships.

SELECTED FIRST-AUTHOR PUBLICATIONS (9 OF 16) & FIELD IMPACT

1. **Zhu, Y.**, et al. 2026, *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over $4.8 < z < 11$*, arXiv:2608.19311.
   Impact: Mapped Lyα visibility for 250 DIVER galaxies over $4.8 < z < 11$, revealing suppressed strong Lyα emission in overdensities—opposite the simplest inside-out reionization expectation—and deriving neutral-fraction ceilings that disfavor an almost completely neutral IGM at $z \sim 8$.
2. **Zhu, Y.**, et al. 2026, *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, *Nature Astronomy*; arXiv:2604.21218.
   Impact: Established metal-enriched baryon cycling in luminous galaxies roughly 500 Myr after the Big Bang through multiphase, blueshifted metal absorption at $7.2 < z < 9.3$.

3. **Zhu, Y.**, et al. 2026, *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, *ApJ*, 1002, 93.
Impact: Located the two clearest $z \sim 5.7$ IGM-transmission regions in galaxy underdensities below half the cosmic mean and delivered the first two-dimensional galaxy–IGM cross-correlation, revealing anisotropic ionization geometry.
4. **Zhu, Y.**, et al. 2025, *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at $z = 6.3$*, *ApJL*, 995, L5.
Impact: Provided evidence that quasar radiation suppresses recent star formation out to approximately 7 comoving Mpc, constraining a roughly 3.1 Myr radiative episode and favoring molecular-gas photodissociation.
5. **Zhu, Y.**, et al. 2025, *SMILES: Potentially Higher Ionizing Photon Production Efficiency in Overdense Regions*, *ApJ*, 986, 18.
Impact: Established the first empirical link between environment and ionizing-photon production efficiency, finding a 2.43-fold higher mean $\xi_{\rm ion}$ in overdense regions and tying environment to reionization topology.
6. **Zhu, Y.**, et al. 2025, *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, *ApJ*, 986, 162.
Impact: Created the first systematic JWST medium-band imaging search for extended line emission and potential outflows, identifying 326 candidates across $1.4 < z < 8.4$ and enabling a scalable census of potential feedback systems.
7. **Zhu, Y.**, et al. 2024, *Damping Wing-Like Features in the Stacked Lyα Forest: Potential Neutral Hydrogen Islands at $z < 6$*, *MNRAS*, 533, L49.
Impact: Detected a damping-wing-like signature around Lyβ dark gaps at mean $z = 5.8$, implying $\langle x_{\rm HI} \rangle \geq 6.1 \pm 3.9\%$ and robustly showing that reionization extended below $z = 6$.
8. **Zhu, Y.**, et al. 2023, *Probing Ultralate Reionization: Direct Measurements of the Mean Free Path over $5 < z < 6$*, *ApJ*, 955, 115.
Impact: Delivered the first direct, multi-bin mapping of the ionizing-photon mean free path across $5 < z < 6$, including first measurements at $z \simeq 5.3$ and 5.6, and showed a fully ionized, relaxed IGM was inconsistent with the data as late as $z = 5.3$.
9. **Zhu, Y.**, et al. 2021, *Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Lyα Forest over $5 < z < 6$*, *ApJ*, 923, 223.
Impact: Established the long-dark-gap fraction as a new reionization diagnostic, tracing its decline from approximately 90% at $z = 6$ to 15% at $z = 5.6$ and finding neutral-island and/or ionizing-background signatures to $z \simeq 5.3$.

RESEARCH & COLLABORATION LEADERSHIP

- **SAPPHIRES Data Lead**: leads data reduction, validation, and delivery for a large JWST pure-parallel imaging and spectroscopy collaboration.
- **SMILES NIRSpec Lead**: leads NIRSpec data reduction, analysis, and the public high-level science-product release.
- **COSMOS-3D Large-Scale Structure Working Group Co-chair**: coordinates environmental and cosmic-web science across the collaboration.

- **Additional major collaborations**: JWST/JADES, JWST/DIVER, ESO VLT XQR-30, and Roman Science Collaboration.

Major-Program Portfolio

- **JWST:** Co-I or data/analysis lead on approximately 2,006 hr of allocated prime, survey, and pure-parallel time across the programs listed below. The total includes approximately 1,600 hr from SAPPHIRES and LAPIS and approximately 406 hr from targeted GO programs.
- **ALMA:** PI of two programs and Co-I on the six programs listed below.
- **ESO/VLT:** Member of Large Programme 112.262L (PI: Emanuele Paolo Farina), *The Cosmic Ecosystem of the First QSOs and Galaxies: a MUSE/XSHOOTER/JWST/ALMA Legacy Survey*; 66 hr with MUSE.

Additional ALMA Co-Investigator Programs

- 2025.1.01067.S (PI: Zhiyuan Ji): *Dust Rising from the ASHES: A Comprehensive ALMA Survey of Remnant Dust in the Earliest Quiescent Galaxies*.
- 2025.1.00805.S (PI: Weizhe Liu): *The Molecular Phase of the Earliest Galaxy-scale Outflows in Quasars at* $z > 6.5$.
- 2025.1.00147.S (PI: Joris Witstok): *Reionization Revealed: ALMA Anatomizes an Enigmatic Ionizing Source within a Galaxy at Redshift 13*.
- 2025.1.00493.S (PI: Manuela Bischetti): *Investigating the Physical Properties and Origin of an Extended [C II] Halo at* $z \sim 6$.
- 2025.1.00215.S (PI: Fengwu Sun): *Dust-Obscured Starbursts in the Most Massive Dark Matter Halos at* $z = 8$*–9: An ALMA+JWST Wide-Field Census*.
- 2024.1.00469.S (PI: Manuela Bischetti): *Investigating the Physical Properties and Origin of an Extended [C II] Halo at* $z \sim 6$.

Selected Major Co-Investigator Programs:

- JWST 12583 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$
- JWST 12577 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 10790 – LAPIS: Large Area Parallel Imaging Survey
- JWST 10311 – Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU
- JWST 8544 – Rest-frame Optical Nebular Emission Lines at Cosmic Dawn: MIRI/LRS Follow-up for JADES-GS-z14-0
- JWST 8060 – JWST Multi-Cycle Deep Transient Survey in GOODS-S
- JWST 8018 – DIVER: Deep Insights into UV Spectroscopy at the Epoch of Reionization
- JWST 7935 – Efficient Measurement of the Emergence Rate of AGN in Legacy Deep Fields
- JWST 7436 – The Last Neutral Islands at the End of Reionization? Characterizing the Nature of the Longest Dark Gaps in IGM Transmission at $z \sim 5.3$
- JWST 7390 – Probing Pair-Instability Supernovae through the Triply-lensed MACS0647-JD at $z = 10.17$

- JWST 7345 – The Dragon Survey: A Direct Probe of the Early Stellar Luminosity Function and Dark Matter through Multi-cycle JWST Multi-cadence Microlensing at $z = 0.73$
- JWST 7336 – Commission a new $R \sim 2500$ "IFU" on JWST: Calibrating second-order spectra of NIRCam/WFSS through the Hubble Ultra Deep Field
- JWST 7335 – Forever Blowing Bubbles: What Powers a 24-kpc Ionized Gas Nebula Around a Normal Galaxy at $z = 6$?

TEACHING & MENTORING

- **Teaching Assistant**, UC Riverside
  - **Summer PhD Comprehensive-Exam Review Course** 2023
    Taught 30 Physics PhD students in an intensive course covering classical mechanics, electromagnetism, thermodynamics, and statistical mechanics.
  - **Undergraduate General Physics Laboratories** 2018–2019
    Led PHYS 2LA (Physics Lab A; 40 students) and PHYS 2LB (Physics Lab B; 40 students); teaching evaluations ranked in the top 20% across campus.
- **Teaching Assistant**, USTC
  - **Introductory C/C++ Programming** 2016
    Taught data structures and algorithms to 160 enrolled students.
- **Zheng Ma**, University of Arizona graduate student 2025–present
  - **Research project:** High-redshift AGN and host-galaxy evolution.
  - **First-author paper:** *A quasar hatching from a buried red phase at* $z = 3.7$ (arXiv:2607.22966).
  - **First-author paper:** *Undermassive Hosts of* $z = 4$*–6 AGN from JWST/NIRCam Image Decomposition* (arXiv:2601.15962; submitted to *ApJ*).
- **Suprabhas Narisetty**, University of Arizona undergraduate 2025–present
  - **Research project:** JWST/NIRSpec study of cosmic-noon galaxies.
  - **Outcome:** First-author manuscript *Strong Balmer Breaks Mark Gradually Quenching Galaxies Below the Star-forming Main Sequence at* $1 < z < 4$ *with JWST/NIRSpec*, prepared for *ApJ*; DOI: 10.5281/zenodo.17772416.
- **G. Hernandez**, UC Riverside undergraduate 2021
  - **Research project:** Evolution of intergalactic-medium effective optical depth.
- **Seyedazim Hashemi**, UC Riverside PhD student — mentored as a peer
  - **Research topic:** Lyα visibility during reionization.
  - **Outcome:** First-author paper *Lyα emission from [O III] emitters near reionization: the role of environment in galaxy Lyα detection*, *MNRAS*, 542, 104 (2025).
- **UCR International Students & Scholars Office** 2021–2023
  - Career and academic mentoring for 14 international graduate students.
- **UCR Graduate Student Mentorship Program** 2020
  - Mentored two junior PhD students across STEM: N. Ahvazi (dark matter and galaxies) and Q. Wu (2D materials).

Courses prepared to teach: introductory astronomy and physics; galaxies and cosmology; observational astronomy; astronomical techniques; computational astrophysics; scientific programming;

and astronomical data analysis.

SELECTED INVITED TALKS & CONFERENCE PRESENTATIONS

* invited

- * Star-forming Galaxies in the Reionization Epoch, Ishigaki, Japan — March 2026
- Galaxy Origins in the JWST Era, Toledo — May 2025
- JWST NIRCam Science Meeting, Biosphere 2 — March 2025
- * NOIRLab Joint Colloquium, **University of Arizona** — February 2025
- * Cosmology Seminar, **Arizona State University** — February 2025
- Lyman-alpha Forest Workshop, **Ohio State University** — October 2024
- JWST MIRI Science Meeting, Biosphere 2 — October 2024
- The First Gigayear(s), Hilo, Hawai‘i — September 2024
- JADES Collaboration Meeting, **University of Copenhagen** — April 2024
- * Galaxy Group Talk, **University of Arizona** — March 2024
- * Galaxy Seminar, **University of Michigan** — November 2023
- Galaxy Formation and Evolution in Southern California — September 2023
- * Kashiwa-Mitaka Meeting (KMM) Seminar, **University of Tokyo** — August 2023
- * Lightning Talk, First Light Conference, **MIT** — June 2023
- Reionisation in the Summer Conference, **MPIA**, Heidelberg — June 2023
- * Galaxy Formation Group, **Northwestern University (CIERA)** — December 2022
- * FLASH Seminar, **UC Santa Cruz** — November 2022
- * Astronomy Lunch Talk, **UC Los Angeles** — October 2022
- * Astro Lunch Talk, **UC Santa Barbara** — September 2022
- Reionization on a Blackboard Workshop, **CCA** — September 2022
- * Special arXiv Coffee, **UC Davis** — May 2022
- * Physics & Astronomy Seminar, **UC Riverside** — April 2022
- * High-z Group Talk, **Tsinghua University** — April 2022
- Reionization and Cosmic Dawn, **UC Berkeley** — March 2022
- European Astronomical Society Annual Meeting (EAS 2021) — July 2021
- Summer All Zoom Epoch of Reionization Conference — June 2021
- * EURECA Seminar, **University of Arizona** — February 2021

PUBLIC OUTREACH & SERVICE

- Invited public talk at Arizona Senior Academy — April 2026
- Session Chair, AAS 247 — January 2026
- Chambliss Judge for student poster competition, AAS 247 — January 2026
- Instructor, *Galactic Adventures*, Flandrau Science Center & Planetarium — June 2025
- Stargazing outreach event, Home Gardens Library (Corona, CA) — October 2023
- Co-organizer, UCR Physics & Astronomy Student Seminar — 2022–2023
- Instructor, UCR Camp Highlander — Summer 2022
- Designed and taught outreach courses for K–12 students — 2022

  - *Multiwavelength Universe*, *Gravity Simulator*
- Virtual Stargazing (UCR & Riverside Astronomical Society): monthly live public outreach on YouTube 2020–2021
- Science fair judge, Riverside County Science and Engineering Fair 2021–2023
- UCR Astronomy Public Outreach: Mercury Transit November 2019

OPEN SCIENCE & COMMUNITY RESOURCES

- **SMILES NIRSpec HLSP (JWST-GTO-1207)**: Lead developer of the public high-level data release including 1D/2D spectra and derived galaxy catalogs.
  GitHub: github.com/ydzhuastro/smiles_dr2; MAST HLSP: doi:10.17909/et3f-zd57
- **Keck/ESI Rest-UV Spectra for Bright Quasars**: Reduced and calibrated spectra for $m_z < 21$ quasars at $z \sim 5.5$, publicly available on Zenodo.
  doi.org/10.5281/zenodo.18677315.
- **Tutorial: Galaxy Morphology $\times$ Machine Learning** February 2025
  Introducing morphology classification and neural network tools for students.
  EURECA Tutorial PDF

SELECTED MEDIA COVERAGE

- "Astronomers Discover a Unique Quasi-Stellar Object-Dusty Star-Forming Galaxy System," **Phys.org**
- "The End of the Cosmic Dawn: Settling a Two-Decade Debate," **SciTechDaily**
- "Can You Explain These Long, Dark Gaps in Your Cosmological Resume?" **AAS Nova**
- "Intergalactic Impacts of Quasars During the Epoch of Reionization" **AAS Nova**
- "Cosmic predators: How supermassive black holes slow star growth in nearby galaxies" **University of Arizona News**
- "Should I Stay or Should I Go? Clumps and Inward Migration" **Astrobites**

TECHNICAL EXPERTISE

- **Observational data and pipelines:** JWST/NIRSpec MSA reduction and calibration; NIRCam imaging and WFSS; PSF matching; slitless-spectroscopy analysis; Keck/ESI and LRIS spectroscopy; ALMA/CASA analysis; public high-level data products.
- **Inference and scientific computing:** Bayesian SED fitting; cosmological simulations; spatial and environmental statistics; machine learning and neural networks; high-performance computing.
- **Programming:** Python, C, Julia, IDL/GDL, and working experience with C++, CUDA, CASA, Gadget-4, GALAXY, and FreeFem++.

YONGDA ZHU - PUBLICATION LIST

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Google Scholar: https://scholar.google.com/citations?user=wDrSZWYAAAAJ

Citation Metrics (as of June 2026):
Total citations: 3500+ | h-index: 35 | First-author citations: 430+ | First-author papers: 16

First-author:

16. **Zhu, Y.**, Fan, X., Keating, L. C., Becker, G. D., Egami, E., Lin, X., Sun, F., Cain, C., Rieke, M. J., Bunker, A. J., Cai, S., D'Eugenio, F., Helton, J. M., Jin, X., Li, M., Ma, Z., Maiolino, R., Rinaldi, P., Willmer, C. N. A., Wu, Y., Wu, Z., and Zhang, J., **2026**. *Low Lyα Visibility in Galaxy Overdensities: Reionization Topology and Neutral-Fraction Ceilings from DIVER over* $4.8 < z < 11$, arXiv e-prints, arXiv:2608.19311. Submitted to *ApJ*.

15. **Zhu, Y.**, Rieke, M. J., Carreira, C., Sun, Y., Ji, Z., Lyu, J., Williams, C. C., Alberts, S., Yuan, F., Zou, Y., Nakazato, Y., Bunker, A. J., Tacchella, S., Sun, F., Rieke, G. H., Egami, E., Chevallard, J., Hainline, K., Ma, Z., Pérez-González, P. G., Rinaldi, P., Rodríguez Del Pino, B., Shields, T., Stone, M., Willmer, C. N. A., Wu, Z., Yue, M., and Zhang, J., **2026**. *PhotoIFU: NIRCam as a Photometric Integral Field Unit for Mapping Feedback in Galaxies*, arXiv e-prints, arXiv:2607.23020.

14. **Zhu, Y.**, Ji, Z., Becker, G. D., Ding, J., Egami, E., Fan, X., Jin, X., Liu, W., Lyu, J., Ma, Z., Narisetty, S., Rieke, G. H., Wu, Y., Yue, M., Zhang, J., and Rieke, M. J., **2026**. *Early metal-enriched baryon cycling before the midpoint of cosmic reionization*, arXiv e-prints, arXiv:2604.21218. Nature Astronomy, accepted in principle

13. **Zhu, Y.**, Becker, G. D., D'Aloisio, A., Endsley, R., Gangolli, N., Cain, C., Mason, C. A., Hashemi, S., and Hong, H., **2026**. *Galaxy Underdensities Host the Clearest Intergalactic Medium Lyα Transmission and Indicate Anisotropic Reionization*, The Astrophysical Journal, 1002, 93

12. **Zhu, Y.**, Rieke, M. J., Ho, L. C., Sun, Y., Rieke, G. H., Yuan, F., Bakx, T. J. L. C., Becker, G. D., Yang, J., Bañados, E., Bischetti, M., Cain, C., Fan, X., Fudamoto, Y., Hashemi, S., Ikeda, R., Ji, Z., Jin, X., Liu, W., Liu, Y., Lyu, J., Ma, H.-X., Takeuchi, T. T., Umehata, H., Wang, F., and Tee, W. L., **2026**. *A Potential Link between Nuclear Winds and Cold Gas Outflows on Kiloparsec Scales in Reionization-era Quasars*, The Astrophysical Journal, 1000, 312.

11. **Zhu, Y.**, Rieke, M. J., Ji, Z., Bunker, A. J., Carreira, C., Danhaive, A. L., Duan, Q., Egami, E., Eisenstein, D. J., Hainline, K., Johnson, B. D., Ma, Z., Puskás, D., Rieke, G. H., Rinaldi, P., Robertson, B., Tacchella, S., Übler, H., Villanueva, N. C., Williams, C. C., Willmer, C. N. A., Wu, Z., and Zhang, J., **2026**. *Clumps in High-redshift Galaxies: Mass Scaling and Radial Trends from JADES*, The Astrophysical Journal, 1000, 303.

10. **Zhu, Y.**, Bonaventura, N., Sun, Y., Rieke, G. H., Alberts, S., Lyu, J., Shivaei, I., Morrison, J., Ji, Z., Egami, E., Helton, J. M., Rieke, M. J., Rinaldi, P., Sun, F., and Willmer, C. N.

A., **2026**. *SMILES Data Release. II. Probing Galaxy Evolution during Cosmic Noon and Beyond with NIRSpec Medium-resolution Spectra*, The Astrophysical Journal, 997, 301.

9. **Zhu, Y.**, Egami, E., Fan, X., Sun, F., Becker, G. D., Cain, C., Chen, H., Eilers, A.-C., Fudamoto, Y., Helton, J. M., Jin, X., Pudoka, M., Bunker, A. J., Cai, Z., Champagne, J. B., Ji, Z., Lin, X., Liu, W., Ma, H.-X., Ma, Z., Maiolino, R., Rieke, G. H., Rieke, M. J., Rinaldi, P., Sun, Y., Tee, W. L., Wang, F., Yang, J., Yue, M., and Zhang, J., **2025**. *Quasar Radiative Feedback May Suppress Galaxy Growth on Intergalactic Scales at z = 6.3*, The Astrophysical Journal, 995, L5.

8. **Zhu, Y.**, Rieke, M. J., Ji, Z., Simmonds, C., Sun, F., Sun, Y., Alberts, S., Bhatawdekar, R., Bunker, A. J., Cargile, P. A., Carniani, S., de Graaff, A., Hainline, K., Helton, J. M., Jones, G. C., Lyu, J., Rieke, G. H., Rinaldi, P., Robertson, B., Scholtz, J., Übler, H., Williams, C. C., and Willmer, C. N. A., **2025**. *A Systematic Search for Galaxies with Extended Emission Lines and Potential Outflows in JADES Medium-band Images*, The Astrophysical Journal, 986, 162.

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